

The γ -flash time base of the n_TOF EAR2 X17 campaign

Calibration from the seven runs with the SiPM-wall divert disabled

n_TOF X17 / DREAM analysis

2026-07-28

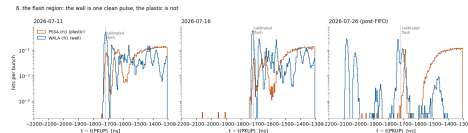
The problem, in one slide

- ▶ The SiPM walls are **blanked** around the γ -flash ($\sim 1\ \mu\text{s}$ gate, N1081B M6 sec. C).
- ▶ So the PSA flash finder tags either
 - ▶ the **gate transient** ($-1190\ \text{ns}$ before 07-22, $-375\ \text{ns}$ after), or
 - ▶ a **clamped leak** at the true flash time, and it **switches between the two from bunch to bunch**.
- ▶ $\Rightarrow$ no per-run offset can repair the stored `tflash`. It has to be replaced.

The fix

$$t_{\text{flash}} = t_{\text{PKUP}} + C, \quad C \simeq -1719\ \text{ns}$$

The pickup is never gated and its finder never fails.



The flash region. Right-hand panel: the wall shows the gate transient at $-2100\ \text{ns}$ and the leak at $-1750\ \text{ns}$.

Finding the runs: a scan of the whole campaign

306 processed runs scanned (224264–224587):

- ▶ **7 runs** with the divert OFF
- ▶ 154 runs with it ON
- ▶ 145 runs with no wall trees at all (before 224345) — unverifiable without a tape reprocess

Signature: all four walls tag the same time within ± 6 ns, and the flash peak is **$\sim 28\,500$ ADC** instead of the ~ 850 ADC gate transient.

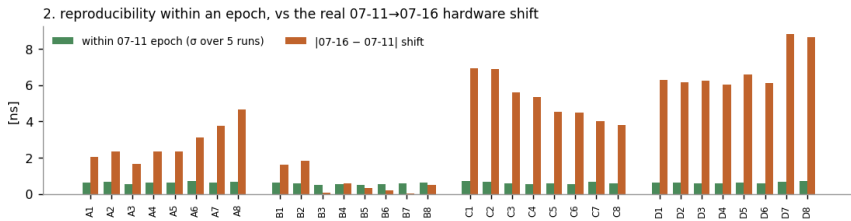
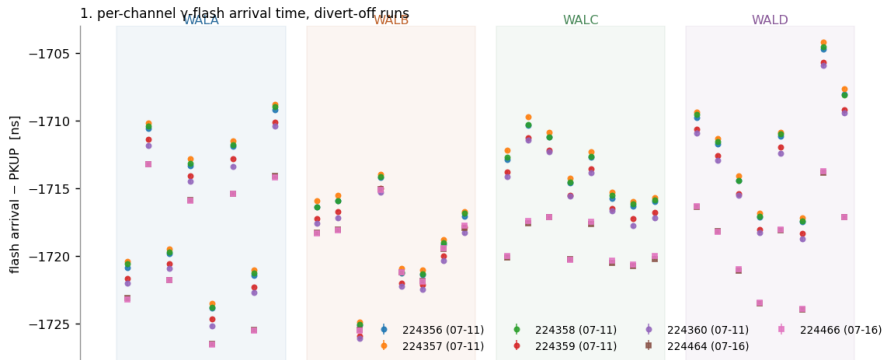
Two independent epochs, five days apart — which is what makes the cross-checks possible.

run	bunches	intensity
224356	3412	dedicated
224357	3286	47/53 ded./par.
224358	3269	mixed
224359	3426	dedicated
224360	549	dedicated
224464	2716	mostly ded.
224466	2073	mostly ded.

07-11 group: N1081B M6 offline during the switch outage.

07-16 pair: during the wall/plastic recalibration.

Per-channel constants



The calibration

wall	all 7	07-11	07-16
WALA	-1717.5	-1716.7	- 1719.5
WALB	-1719.3	-1719.1	- 1719.7
WALC	-1715.5	-1714.0	- 1719.2
WALD	-1714.1	-1712.2	- 1719.0
all 32	-1716.6	-1715.5	- 1719.3

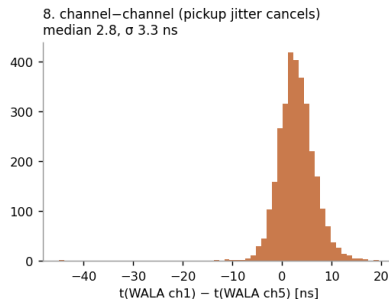
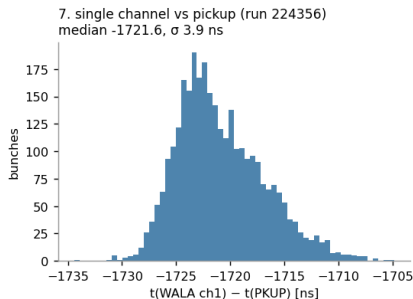
Per-wall means, for orientation only. The calibration is **per channel** — 32 constants, 07-16 epoch, runs ≥ 224400 .

scheme	rms resid.	max
one constant, all 32	3.43	7.18
one per wall	3.42	7.07
per channel	0	0

Wall means agree to 0.7 ns; channels within a wall differ by up to 13.3 ns $\Rightarrow$ grouping by detector captures **nothing**.

uncertainty	[ns]
run-to-run, within epoch	0.5
time base over campaign (rms)	0.68
transport to end of campaign	1.2
07-11 vs 07-16 epoch	3.9
intensity, par. $\rightarrow$ ded.	5.0
single bunch, one channel	3.5
single bunch, 32-ch average	2.5

Timing resolution: what is in the 3.5 ns



Decomposing the 32-channel matrix, per bunch:

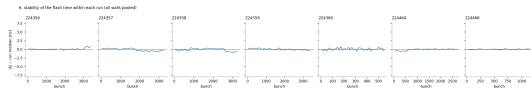
- ▶ single channel vs pickup: 3.5 ns
- ▶ common to all 32 channels: 2.5 ns
- ▶ independent per channel: 2.5 ns

The independent part averages away over the array; the common part does not. The pickup contributes only ~ 1 ns, so the common 2.5 ns is a **genuine bunch-to-bunch variation of the flash arrival**.

07-16 runs are better than 07-11 (2.7 vs 3.9 ns) — again the equalisation.

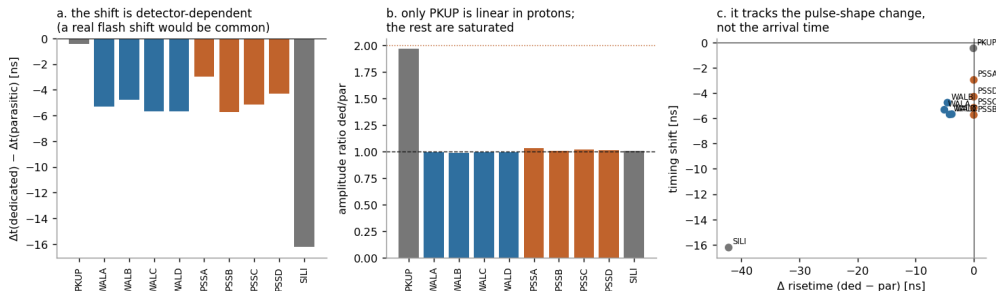
The reference: why the pickup

- ▶ per-bunch spread **0.83 ns** (dedicated), 1.42 ns (parasitic)
- ▶ usable in **98.4 %** of bunches (the 1.6 % failures are all parasitic)
- ▶ amplitude strictly **linear in protons**: ratio 1.97 for a factor 2
- ▶ timing moves by only -0.45 ns between intensity classes
- ▶ its flash finder **never fails** — unlike WAL/PSS/LIQ



Within a full three-hour run the flash time is flat to 0.25–0.39 ns bin-to-bin, 1.4 ns peak-to-peak.

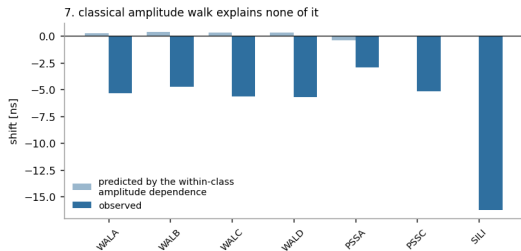
Question 1: is the intensity dependence real?



The wall constant moves by -5.0 ns from parasitic to dedicated pulses. But:

- ▶ pickup 0; walls -5 ; plastics $-3 \dots -6$; **liquids** $-4.1 \dots -4.9$; silicon -16 ns. The scintillators cluster where “more light $\rightarrow$ earlier crossing” predicts; a real change of the flash *arrival* would have to move the pickup too.
- ▶ only the pickup is linear in protons; everything else is saturated (amplitude ratio $\simeq 1$)
- ▶ the shift follows the **risetime change**: walls $20.5 \rightarrow 15.4$ ns, silicon $98 \rightarrow 56$ ns

Question 1: and it is not classical amplitude walk



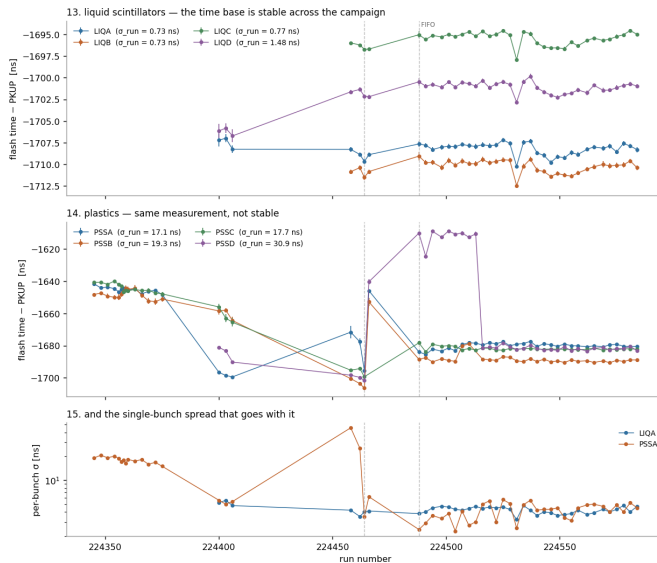
- ▶ Measure the amplitude→time slope *within* one intensity class.
- ▶ Extrapolate it to the observed amplitude change *between* classes.
- ▶ Prediction: +0.3 ns (walls, **wrong sign**), < 0.4 ns elsewhere.
- ▶ Observed: -5 to -16 ns.

Mechanism

The front end saturates: a bigger flash cannot make a taller pulse, so it reaches the clipping level **earlier** and stays there longer (FWHM 700 → 2000 ns). Any edge estimator fires early.

⇒ carry it in the calibration; do *not* read it as physics.

Question 2: is the time base stable across the campaign?



LIQ and PSS are **never gated** so they see the flash in every run — the only way to test stability where the walls

Liquids: stable

- ▶ $\sigma_{\text{run}} = 0.73, 0.73, 0.77, 1.48 \text{ ns}$ (A–D) over 36–39 runs, 224400–224584
- ▶ the four cells **move together**: within-run spread of deviations 0.22 ns , LIQA–LIQC correlation $+0.92$
- ▶ common-mode wander **0.68 ns rms** , worst single run -2.2 ns (224531); at *fixed* intensity it tightens to **$0.41\text{--}0.76 \text{ ns}$** (LIQA/B/C)
- ▶ the 07-16 divert-off runs sit $+0.9 \dots +1.2 \text{ ns}$ from the late-campaign mean $\Rightarrow$ **transport good to 1.2 ns**

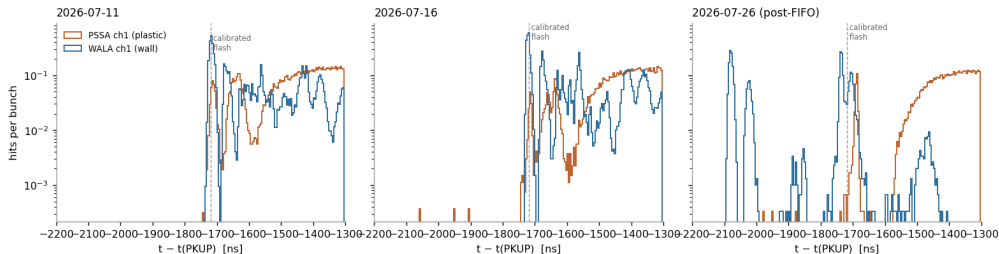
Independent check in gated runs

- ▶ the clamped wall leak lands within **6.2 ns** of the calibration (worst case 9.2 ns), on all 32 channels
- ▶ in both 224362 (07-11 era) and 224572 (07-26, 15 days later)
- ▶ the gate transient sits at -1190 and -375 ns — reproducing the $+800 \text{ ns}$ delay change of 07-22

The wall calibration measured on 07-16 carries to the end of the campaign at the $\sim 1 \text{ ns}$ level.

Question 3: the plastics — anatomy

8. the flash region: the wall is one clean pulse, the plastic is not



The wall gives **one clean pulse**. The plastic response is spread over ~ 100 ns and sits on a continuum of fragments: **25–30 hits per bunch** in the $\pm 1 \mu\text{s}$ window, against 12 for the walls.

Question 3: the plastics — three failure modes

epoch	family	found	σ [ns]	amplitude	sat. flag	FWHM	n_{hit}/b
2026-07-11	WAL	100 %	4.1	28521	0 %	1527	12.4
2026-07-11	PSS	67 %	19.1	23193	0 %	11	38.2
2026-07-16	WAL	100 %	2.3	28560	0 %	2602	12.5
2026-07-16	PSS	64 %	5.8	34571	0 %	43	26.0
2026-07-16	LIQ	100 %	4.1	63931	80 %	103	43.6
2026-07-26	WAL	100 %	8.3	19650	6 %	57	11.7
2026-07-26	PSS	42 %	2.5	55114	27 %	76	24.3
2026-07-26	LIQ	100 %	4.5	63921	79 %	102	44.1

1. **Yield.** A flash hit exists in only 64–67 % of bunches, 42 % after the FIFO fan-in. In the missing bunches the PSA reconstructs *no hit at all* before ~ -1600 ns — verified not to be a selection artefact.
2. **Amplitude-dependent bias** up to 17 ns, $\sigma = 7\text{--}15$ ns (pre-FIFO).
3. **Instability:** the constant moves tens of ns across the campaign ($\sigma_{\text{run}} = 17\text{--}31$ ns).

Question 3: it is not “saturation breaks the PSA”

The **liquid scintillators are the control**: same digitiser, same PSA, different front end.

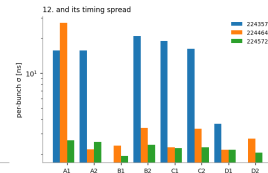
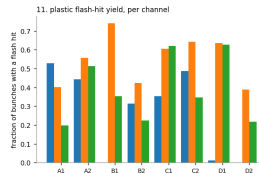
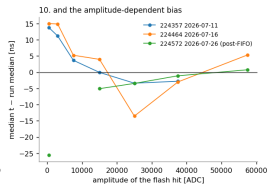
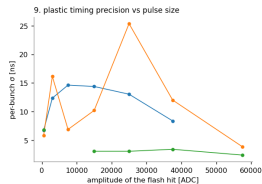
- ▶ LIQ flash hits are **more** saturated than the plastics': sat. flag **79–80 %** vs 27 %, amplitude 63 900 vs 55 100
- ▶ highest hit multiplicity of all: **44**/bunch
- ▶ and yet the PSA finds the flash in **100 %** of bunches, at $\sigma = 4.1\text{--}4.5\text{ ns}$

So the plastic drop-out is. . .

specific to the plastic channel — its PSA configuration and/or its pulse shape (PSS flash FWHM 43–76 ns and enormous area) — **not** a generic consequence of a railed pulse.

⇒ plausibly **fixable in the UserInput**, which is exactly what the reprocessing effort is for.

Question 3: precision vs pulse size, and per channel



The two regimes trade off: pre-FIFO the hit is usually found but badly timed; post-FIFO it is precisely timed (2.4–3.4 ns) but usually missing. PSSB ch2 is defective throughout (amplitude 9 778 vs ~30 000). $\Rightarrow$ take plastic constants **per run**, never per epoch.

Update: the plastics are fixed by the reprocessing

The parallel UserInput work reprocessed run 224572. Same measurement on v4_walshapes:

	official	reprocessed
PSS flash hit found	42 %	100 %
per-bunch σ	2.5 ns	3.0–4.6 ns
LIQ found	100 %	100 %

The single change responsible is the **G-FLASH threshold** ($50 \rightarrow 2000/10^4$): v1_flash alone, elimination untouched, already gives 100 %.

So the missing hits were. . .

a **downstream symptom of the flash-finder bug**, not an independent plastic defect. With the flash mis-located at ~ 130 ns (fitting noise), the pulse recognition never reconstructs the real pulse at $11.63 \mu\text{s}$.

Elimination is *not* the gate: widening it (v2_elim) changes the yield not at all, and the surviving flash hits have area/amp $\simeq 72$ – 76 , above even the widened window.

$\Rightarrow$ § “plastics unusable” applies to the *official* files only.

Independent cross-check: the two routes agree

tree	reprocessed	this work	diff
WALA	-1717.82	-1719.46	+1.64
WALB	-1718.07	-1719.69	+1.62
WALC	-1718.05	-1719.23	+1.18
WALD	-1722.29	-1719.01	-3.28
		rms	2.07
LIQA	-1708.38	-1708.22	-0.16
LIQB	-1710.78	-1710.30	-0.48
LIQC	-1695.73	-1695.56	-0.17
LIQD	-1701.27	-1701.56	+0.29
		rms	0.27

A **corrected flash finder** run on 224572 — a *gated* run, ten days after the nearest calibration run, where the walls see only the clamped leak.

Walls agree to **2.1 ns** rms, liquids to **0.27 ns** rms — two different processings of the same run agreeing to a quarter of a nanosecond.

The two routes share no assumption beyond the pickup itself.

1. **Walls:** per-channel C_2026_07_16 (**all 32**) for runs ≥ 224400 , C_2026_07_11 before — never one constant per wall. Add -5 ns if the pulse type differs from the calibration one.
2. **Liquids:** per-run value or campaign mean — they agree to $\sim 1\text{ ns}$.
3. **Plastics:** per-run value only.
4. **Never** use the stored `tf1ash` of WAL/PSS/LIQ *from the official files*; in reprocessed files it is sound, but PKUP + C is still the recommended time base.
5. If a bunch has no usable pickup pulse (1.6 %, all parasitic), drop the bunch.

Limitations

Amplitude is unusable in these runs (front end saturated) — energy scale still from the ^{88}Y runs 224476–479. The 3.9 ns epoch shift shows wall timing *can* move; nothing rules out a further move after 07-16 beyond the liquid evidence. Runs before 224345 unverified. Raw data of all seven runs has aged off EOS disk.

data/flash_timing_calibration.json	the calibration, per channel and epoch
data/per_channel_flash_timing.csv	280 measurements behind it
data/divert_state_by_run.csv	the 306-run divert scan
data/plastic_liq_flash_by_run.csv	the time-base monitor
data/plastic_pathology.csv	the plastic study
data/resolution_decomposition.json	jitter split, per run
figures/, scripts/, latex/	plots, code, this deck

$$t_{\text{flash}} = t_{\text{PKUP}} + C, \quad C \simeq -1719 \text{ ns}$$